

Statement of Work (SOW)

for

Single Mode (SM) Fiber Optic Cable (FOC)
DS – Manhole Distribution System (MHDS)/FOC US Army Adelphi Laboratory Center
Building 205 to Arnold Engineering Development Complex (AEDC) Building 405
White Oak MD

March 19, 2026

Install Location Address
Arnold Engineering Development Complex (AEDC) White Oak
Operating Location B
9 Sperry Rd
Building 405 Room 1319
Silver Spring, MD 20903

United States Army Adelphi Laboratory Center
2800 Powder Mill Rd
Building 205 Room 1D-028B
Adelphi, MD 20783

AEDC Communications Office of Responsibility
AEDC/TSDI
100 Kindel Drive
Room B213B
Arnold Air Force Base (AAFB), Tennessee 37389-2212

IRA-26-1277

Table of Contents

1.0	SCOPE.	4
2.0	REQUIREMENTS.	5
2.1	GENERAL REQUIREMENTS.	7
2.1.1	Safety Requirements.	7
2.1.1.1	Site Coordination.....	7
2.1.1.2	Confined Space.	7
2.1.1.3	Accident/Incident Reporting and Investigation.....	7
2.1.1.4	Work Area(s).....	7
2.1.1.5	Traffic Control.....	7
2.1.2	Security Requirements.	8
2.1.2.1	Security Clearances.	8
2.1.2.2	Operational Security (OPSEC).....	8
2.1.3	Environmental Management.	8
2.1.4	Permits.....	8
2.1.5	Integrated Product Team (IPT).	9
2.1.6	Weekly Status Reports.	9
2.1.7	Quality Assurance.	9
2.1.8	Contractor Personnel.	9
2.1.8.1	Project Management.....	9
2.1.8.2	Site Point of Contact (POC).	9
2.1.8.3	Personnel Requirements.	9
2.1.9	Warranty.....	9
2.1.10	Manuals and Practices.	10
2.2	SPECIFIC REQUIREMENTS.....	10
2.2.1	Outside Plant Requirements.	10
2.2.1.1	Existing Maintenance Holes.....	10
2.2.1.2	Measurements.....	10
2.2.1.3	Cable Racks and Cable Rack Supports.	10
2.2.1.4	Labeling.....	10
2.2.1.4.1	Cable Tags.....	10
2.2.1.4.2	FODP Patch Panel Labeling Format.	11
2.2.1.5	FOC Terminations and Splicing.....	12
2.2.1.6	FOC Maintenance Loop.	12
2.2.1.7	Outside Plant Installation.	12
2.2.1.7.1.1	Handhole and Conduit System.....	12
2.2.1.7.1.2	Composition.	13
2.2.1.7.1.3	Installation.	14
2.2.1.7.1.4	Paved Surface Crossings.	14
2.2.1.7.1.5	Conduit Protection.....	15
2.2.1.7.1.6	Conduit Bends, Sealing and Mandrelling.	15
2.2.1.7.1.7	Building Entrances, Pull Boxes, and NEMA 4 Enclosures.....	15
2.2.1.7.1.8	Entrance Conduits into Existing Maintenance holes and Handholes.....	16
2.2.1.7.1.9	Pull Tape.	16
2.2.1.7.1.10	Utility Separation.....	16
2.2.1.7.1.11	Spacers and Tracer Wire.....	17

2.2.1.7.2	Closure for Underground Optical Fiber Cables.	17
2.2.1.7.3	Fiber Optic Distribution Panels (FODPs).	17
2.2.1.7.4	Fiber Optic Cable Installation.	Error! Bookmark not defined.
2.2.1.8	Testing.	17
2.2.1.8.1	Fiber Optic Cable Tests.	18
2.2.1.8.1.1	Optical Attenuation.	18
2.2.1.8.1.2	Fiber Testing.	18
2.2.2	Site Restoration/Debris Removal.	18
2.2.3	Service Outages.	19
2.2.4	National Pollution Discharge Elimination System (NPDES).	19
2.2.5	Underground Utilities Markings.	19
2.2.6	Installation Schedules.	19
2.2.7	As-Built Drawings.	20
2.2.8	Installation Test Plan.	21
2.2.9	Final Acceptance.	21
2.2.12	Deliverable Summary.	21
3.0	GENERAL INFORMATION	21
3.2	Place of Performance	21
3.3	Hours of Operation.	22
3.4	Holidays/Down Days	22
3.5	Base Support	22
4.0	APPENDICES.	23

1.0 SCOPE.

This Statement of Work (SOW) defines the requirements for the Contractor to provide a turnkey, design, engineer, furnish, install, and test new fiber optic cable from the US Army Adelphi Laboratory Center Building 205 Room 1D-028B to the AEDC Operating Location B, White Oak Building 405 Room 1310.

The following details the requirements for this work:

- The Contractor Shall conduct an Independent Site Visit to determine final design, and cost and ensure all activities can be accomplished at the site through the existing MH Base Wide Duct System prior to submission of their proposal.
- The Contractor Shall install Single Mode (48SM) microfiber and Microduct innerduct (one-7-way) system from Building 205 to Building 405. Install 48 strands of SM Microfiber FOC, and 7-Way/cell Microduct Innerduct from Building 205 Room 1D-028B to Building 405 Room 1310. Inside facility conduit will be required from the outside wall penetration to the existing Cabinet in building 205 and building 405.
- The Contractor shall furnish, install, and test all associated equipment required to provide a turnkey, design, engineering and installation solution to establish the fiber connectivity as indicated in this SOW.
- All equipment, supplies, or materials provided shall be new and not refurbished. No equipment, supplies, and material will be provided by the government.
- The Contractor shall provide a proposal to the government The Contractor's proposal will be reviewed for technical acceptability in accordance with the following factors:
- The Contractor's proposal shall contain sufficient design drawing detail showing the projected path, Butterfly drawings and detailed explanation of how the requirements of this SOW will be completed.
- The Contractor's proposal shall sufficiently detail all costs associated with the requirements of this SOW.
- The 7-Way/cell Microduct shall be installed through the existing Base Wide Duct System, if there is not sufficient room for the 7-Way/cell Microduct, alternative size microduct shall be proposed.
- All innerducts shall be installed IAW the Host Base requirements within the Manholes.

SITE DESCRIPTION

The AEDC White Oak (Operating Location B) is located on the Federal Research Center Campus, sharing the site with the Food and Drug Administration in White Oak MD. Due to the Base Realignment and Closure in the 90s, what was originally a Naval base was broken up. The side where the DeMarc is located was taken over by the US Army and Operating Location B was sold to the USAF. The Host Base providing telecommunication and internet services is the US Army Adelphi Laboratory Center, operationally under the Aberdeen Proving Grounds is a major stakeholder and owns building 205 that the fiber will originate at. There are multiple

Environmental agencies associated with the White Oak area that will require coordinating any infrastructure installs at the Federal Research Center at White Oak.

2.0 REQUIREMENTS.

The Contractor proposal shall address all costs and provide a detail of all work, materials, and equipment for each requirement below.

Any distances provided in this SOW are approximations and shall NOT be used for ordering materials such as cable, Microduct, etc., or determining duct lengths. The Contractor shall be responsible for a turnkey design, engineering solution, and installation of all SM cable, handholes, seven-way (7-Way) micro duct and ancillary supporting equipment required for this project. If there is not sufficient space in the existing Base Wide Duct System to run the 7-Way microduct, alternative options shall be proposed.

CLIN 0001 – Site Visit Assessment White Oak MD

CLIN 0002 – Fiber Optic Cable Installation Services (Labor) Building 205 to Building 405 DeMarc.

CLIN 0003 – Fiber Optic Cable (Hardware) Building 205 DeMarc to Building 405

CLIN 0004 – Equipment Rental for Fiber Optic Cable Installation Services Building 205 to Building 405.

Building 205 Room 1D-028B Infrastructure Installation:

- Install one new 4U FODP in existing equipment cabinet in Building 205 DeMarc Room 1D-028B to accommodate the 48 Strand SM fiber.
- Install new 7-Way/Cell Microduct Innerduct approximately twelve-thousand four hundred and eight feet (12,408) from Building 205 CER to Building 405 CER.
- Install new external Galvanized Steel NEMA pullbox at FOC entry location on facility if existing NEMA Pullbox cannot be used.
- Install new four-inch (4”) LB galvanized steel conduit into Building 205 NEMA Box as required.
- Install and terminate 48 Strand Single Mode (SM) Microfiber.
- Terminate, Test and provide test reports of all terminated fiber with LC connectors.
- Install new conduit inside building 205 to accommodate the Microduct Innerduct from the Outside Wall penetration to CER room 1D-028B, approximately 100 feet.

Building 405 Room 1310 Infrastructure Installation:

- Install new one-way four-inch (4") external Galvanized Steel NEMA pullbox at FOC entry location of facility if existing NEMA Pullbox cannot be used.
- Install new four-inch (4") LB galvanized steel conduit into Building 405 NEMA Box as determined by site visit.
- Install new conduit as required from outside wall penetration to Room 1310 CER, approximately 20 feet.
- Install new 7-Way Microduct to existing equipment Cabinet.
- Install one new FODP in existing equipment cabinet in Building 405 Room 1310.
- Install and terminate 48 Strand Single Mode (SM) Microfiber.
- Terminate, Test and provide test reports of all terminated fiber with LC connectors.

Single Mode (SM) Micro Fiber Optic Cable (FOC):

- The Contractor shall install and terminate approximately Twelve-thousand four-hundred and eight feet (12,408 ft) of new 48 strand SM MicroFiber from Building 205 Room 1D-028B through new existing Base Wide Duct System through new Microduct Innerduct to Building 405 Room 1310, see Attachment 1 fiber path drawing.
- The Contractor shall use 48 Strand Single Mode Outdoor Rated Microfiber Optic Cable (SM FOC)
- Specification: Contractor shall provide Specification Sheet of all material for Government Approval prior to ordering material.
- The Contractor shall route and terminate all SM FOC with Lucent Connector (LCs).

Manhole Distribution System (MHDS):

- The contractor shall use existing Base Wide Duct System, to install the new Innerduct and Microfiber under close coordination with the Host Base personnel.

Microduct and Handhole Installation:

- The Contractor shall install approximately 12,408 feet new Seven-cell (7-way) Microduct through existing Handholes From Building 205 Room 1D-028D, location of DeMarc to Building 405 Room 1310.
- The Contractor shall install one tracer wire approximately 12,408 feet.
- The Contractor shall install 7-Way/Cell Microduct sufficiently sized for installing new 48 strand Microfiber with a Tracer Wire.

The below material list is an estimate only and is provided for the approximate use as the Contractor sees fit. The Contractor shall conduct a survey to identify technical solution and material requirements.

Building 205 to Building 405 DeMarc Room 1310 Approximate Materials

Material	QTY	Note
4U FODP	2	Contractor survey shall determine final solution
7-Way/cell Microduct	12,408 ft	Contractor survey shall determine final solution
48 SM FOC (MicroFiber)	13,000	Contractor survey shall determine final solution
4 inch Panduit	300 ft	Contractor survey shall determine final solution

2.1 GENERAL REQUIREMENTS.

2.1.1 Safety Requirements.

2.1.1.1 Site Coordination.

The Contractor shall meet with the Safety Officers of all involved parties immediately upon arrival on-site for review of the specific safety requirements prior to installation.

2.1.1.2 Confined Space.

The Contractors and any subcontractors entering spaces on AEDC White Oak and ALC are responsible for the safety of their personnel, any subcontractor personnel, and for their own confined space program as outlined in AFI 91-203 and IAW Occupational Safety and Health Administration (OSHA). Before entering any confined space at ALC, FDA, and AEDC White Oak, the Contractors and any subcontractors must present valid documentation of confined space training in advance for proper validation prior to entering any confined space.

2.1.1.3 Accident/Incident Reporting and Investigation.

The Contractor shall record and report all available facts relating to each instance of injury to either Contractor or Government personnel to the Site Safety Office. The Contractor shall secure the scene of any accident and wreckage until released by the accident investigative authority through the Contracting Officer. If the Government elects to investigate an incident, the Contractor shall cooperate fully and assist the Government personnel until the investigation is completed.

2.1.1.4 Work Area(s).

At day's end, the Contractor shall remove all debris and surplus materials from the workplace. Safety barriers shall be in place to protect unfinished work site at the end of each day. All open holes or trenches shall be completely enclosed by flexible orange safety fencing, or other safety barriers, at the end of the workday. On work sites outside the base perimeter, the Contractor shall coordinate with and abide by the requirements of the appropriate local, state, and federal agencies to ensure work areas are safe at day's end.

2.1.1.5 Traffic Control.

In the event base vehicular traffic is to be disrupted by trenching or horizontal directional boring,

the Contractor shall notify the Program Manager no-later-than (NLT) ten (10) calendar days in advance to inform base Security Forces and emergency services personnel of the planned disruptions. On work sites outside the base perimeter, the Contractor shall coordinate any traffic disruptions with local authorities.

2.1.2 Security Requirements.

2.1.2.1 Security Clearances.

The Contractor is required to work in secure/controlled areas. The Contractor shall process a Visitor Request Email to the AEDC White Oak PM to coordinate a Visitor Badge for access to any areas within the fence line at ALC and AEDC White Oak. The Contractor shall provide any necessary personally identifiable information (PII) requested by the Site Security Manager for Contractor and subcontractor employees performing work on this SOW at White Oak MD. During project implementation, the Contractor shall coordinate access to secure areas at least twenty-four (24) hours in advance with the AEDC White Oak PM.

2.1.2.2 Operational Security (OPSEC).

The Contractor shall protect and guard against loss or duplication of any and all provided network infrastructure drawings (MHDS, handhole locations, fiber paths, etc.). The Contractor shall take appropriate measures to protect detailed information pertaining to this effort, including appropriate marking of documents as “CUI – Controlled Unclassified Information” and ensuring limited distribution of documents and schematics/drawings to only those individuals with a valid need to know. In accordance with AFI 10-701, Operations Security (OPSEC), the Contractor shall develop an OPSEC plan to ensure the protection of all CUI data, either furnished by the Government, or produced by the Contractor.

2.1.3 Environmental Management.

The Contractor shall comply with all federal, state, and local laws and regulations, Air Force policies, instructions, and plans. The Contractor and its subcontractors shall maintain an awareness of changing environmental regulatory requirements to avoid environmental deficiencies for activities at ALC and AEDC White Oak.

2.1.4 Permits.

The Contractor shall complete and process all permits required to perform under this Contract and SOW

- The Contractor shall obtain a Master Work Permit (MWP) GC-1732 Form Sections I-III coordinated and approved by the MWP issuing official, which should be coordinated through the AEDC White Oak PM prior to any facility or maintenance modification.
- The Contractor shall obtain a MWP GC-1732 Form Sections IV-V (Dig Permit) coordinated and approved by the dig permit issuing officials, which shall be coordinated through the Program Manager at least fourteen (14) calendar days in advance of any need date.
- The Contractor shall coordinate Air Force Form 1024s, Confined Spaces Entry Permit, with the Safety Office seven (7) calendar days in advance. The Contractor shall be prepared to provide proof of Confined Space Entry Safety training program, along with the AF Form 1024, to the site safety office POC.

- The site Program Manager working with the Host Base will provide the appropriate form(s) in a timely manner upon request by the Contractor.

2.1.5 Integrated Product Team (IPT).

The Contractor shall chair a weekly Integrated Product Team (IPT) meeting which includes Contractor representatives, the designated Program Manager, and other base personnel as requested or as necessary to discuss project progress, problems being encountered, and other information necessary/beneficial to ensure success and timely completion of contract requirements. The Contractor shall provide an agenda and a worldwide “Meet Me” teleconference capability for the duration of the project. The Contractor shall provide a weekly report (soft copy) indicating progress/status, any system or outside plant deficiencies/discrepancies found and actions to correct them.

2.1.6 Weekly Status Reports.

The Contractor shall prepare a Weekly Status Report and distribute to IPT members. The purpose of the report is to inform IPT members of project progress, problems being encountered, and other topics necessary/beneficial to ensure success and timely completion of the contract requirements. The Status Report and meeting agenda may be combined as long as the resulting report contains all the required elements and contains sufficient detail to serve as project record.

2.1.7 Quality Assurance.

See 2.1.5 above.

2.1.8 Contractor Personnel.

2.1.8.1 Project Management.

The Contractor shall employ a Project Manager (PM) and alternate(s) responsible for Contract performance and continuity. The Contractor shall identify the Contract Project Manager’s or alternate’s authority to act for the Contractor relating to daily contract operation.

2.1.8.2 Site Point of Contact (POC).

The Contractor shall designate the Contractor’s on-site team leader and alternate(s) as the site point of contact (POC) for individual projects in a site visit request letter. The Site POC or alternate(s) shall be on-site during duty hours until project completion. The Site POC shall be the interface for all work site communications with the Government, including quality, safety, and discrepancy matters.

2.1.8.3 Personnel Requirements.

The Contractor’s Project Manager, Site POC, and respective alternate(s) shall be able to read, write, speak, and understand English.

2.1.9 Warranty.

The Contractor shall provide a standard commercial warranty of one (1) year or more for all materials and work furnished or performed under this Contract. The warranty period shall start from the date of physical completion/project acceptance in Wide Area Workflow (WAWF). The Contractor shall provide written procedures and required information for warranty services at or

prior to site acceptance.

2.1.10 Manuals and Practices.

The Contractor shall provide to the Program Manager the latest version of operation, installation, and maintenance manuals and practices/users' guides for each system installed as provided by the original manufacturer with all new equipment.

2.2 SPECIFIC REQUIREMENTS.

The Contractor shall provide all equipment, tools, materials, supplies, transportation, labor, supervision, management, etc. necessary to meet the requirements as stated in this SOW. The Contractor shall comply with the current Telecommunications Industry Association (TIA/EIA) and Building Industry Consulting Services International (BICSI) Telecommunications Distribution Methods Manual, telecommunication installation and testing commercial standard, and base installation standards.

2.2.1 Outside Plant Requirements.

2.2.1.1 Existing Maintenance Holes. N/A

2.2.1.2 Measurements.

Any distances provided in this SOW are approximations and shall NOT be used for ordering materials such as cable, innerduct, etc., or determining duct lengths. The Contractor shall be responsible for a turnkey design, engineering solution, and installation of all SM cable, seven-way (7-Way) Microduct and ancillary supporting equipment required for this project.

2.2.1.3 Cable Racks and Cable Rack Supports.

The Contractor shall install cable racks if none are available in the existing Manholes. Splices shall not be supported by the cables that enter each end of the splice case. The splices shall be supported by cable hooks under the splice case. Telecommunications industry standard cable hooks of the appropriate length shall be provided to support cables and splice cases. The cable hooks shall be secured using cable rack locking clips. All cables shall be supported using racking clips, cable racks, and cable hooks.

2.2.1.4 Labeling.

The Contractor shall label all equipment and cables they install in accordance with TIA-606 and as required by the site PM.

2.2.1.4.1 Cable Tags.

All tags shall be permanently labeled, easily visible, and corrosion resistant. Cable tags shall be installed in all handholes and fiber optic distribution panel (FODP) locations. When cables pass through Manholes, the Contractor shall add a tag on the cable, approximately two feet (2') from each duct entrance and at each splice location. Information on the cable tag shall identify cable by size, type, cable number and count. The same information shall appear on the Contractor's completed as-built-drawings.

a. Example: **Outside Plant Labeling Format**

TO: Bldg|Rm|FPI|Patch Panel|Port|Type|

Bldg No.
Room No.
Fixed Point ID (FPI) – based on facility/room/rack drawing
Patch Panel ID
Module No.
Port No.
Cable Type

FROM: Same info format as above

CCSD/Circuit ID No.|Circuit Bandwidth

Example:

100 |B202| FPI-A13/D8|CA901234| 1-2 | SMF

251 |313| FPI-A18/A1|CA901234| 1-2 | SMF

7KLY|50 Mb

2.2.1.4.2 FODP Patch Panel Labeling Format.

The fiber optic distribution panel (FODP) shall be stenciled/marked with black ink, black paint, or adhesive backed decals in letters and numbers. If the manufacturer has not identified the sequence in which ports on FODP's pigtail modules are counted, the Contractor shall provide designation labels/strips to identify the sequence in which they are counted. Each splice tray shall be marked to identify the fiber count contained in the splice tray. The marking shall identify the FODP by number, building number, cable number, count, etc.

FODP example:

a. **Patch Panel Labeling Format**

**TO: FPI | Patch Panel ID | From Bldg | From RM | To Bldg | To RM |
Count | Type**

Fixed Point ID (FPI) – based on facility/room/rack drawing
Patch Panel ID
Bldg No.
Room No.
Count – No of cables
Type cable (Copper, Single Mode, Multimode)

FROM: Same info format as above

Example:

FPI-G18 | CA901234 | 100 | B202 | 36 | SMF (example only)
FPI-B12 | CA902345 | 251 | 313 | 36 | SMF

New Ducts.

New ducts shall be permanently labeled on the wall of each building/handhole (handhole) indicating the connecting building/maintenance hole/handhole at the other end of the duct (for example, "To handhole-306H"). The same information shall appear on the Contractor's completed as-built-drawings.

2.2.1.5 FOC Terminations and Splicing.

All FOC entering a building will terminate in the telecommunications room (TR) into the existing floor mounted cabinet and new FOC patch panel with LC connectors. All connectors will be fuse-able link type with no mechanical connectors used.

Fiber optic cables shall be terminated using LC connectors at the FODP by direct connection. The Contractor shall install connector directly onto the incoming optical fiber using fusion connections (NO MECHANICALS). The Contractor shall identify whether or not there is some practical reason for an intermediate splice in the cable at some handhole between the cable end points (no Direct Buried or BD CAN Splices will be permitted) and consist of fusion splices in a high quality re-enterable splice case fiber optic closure. A coil of seventy-five feet (75') of cable shall be provided on each cable entering or leaving a splice case in a handhole.

2.2.1.6 FOC Maintenance Loop.

The Contractor shall allocate a minimum of seventy-five feet (75') of fiber optic cable at every Manhole/handhole location and every 7-Way Microduct joining buried point, starting/ending in the first maintenance hole outside each end building (*i.e.*, the first maintenance hole outside of each building will contain a 75 feet maintenance loop). All maintenance loops installed within maintenance holes must be supported by two cable hooks. Cable hooks are to be positioned so the highest one supports the underside of the top of the coil and the bottom hook supports the underside of the bottom of the coil. The maintenance loop shall have a cable tag. The same information shall appear on the Contractor's completed as-built drawings.

2.2.1.7 Outside Plant Installation.

The Contractor shall design and install outside plant telecommunications infrastructure in accordance with TIA-758. Each cable installation and cut over shall be coordinated with the AF Designated Program Manager. The sequence of installation is at the Contractor's discretion. See Attachment 1 for reference and estimated distances of existing and proposed cables routes.

2.2.1.7.1 Conduit Systems and Inner-ducts Installation.

As part of a turnkey design, engineering, and installation, the Contractor shall provide design drawings for Government approval. Approved designs shall include any handhole and conduit IAW industry standards.

2.2.1.7.1.1 Handhole and Conduit System

1. ALL material shall be submitted by the Contractor to the AEDC White Oak Program

Manager for this project as a submittal for approval prior to purchasing and installation.

2. Handhole configuration and requirements are further defined in below paragraphs and bullets.
3. The Contractor shall install 7-Way/cell microduct innerduct from Manhole/handhole to building demarcation with one of cell being locatable.
4. Concrete Encased Conduit. N/A

2.2.1.7.1.2 Composition.

MicroDucts

New Microduct Innerduct shall be corrosion resistant and comply with the latest standards for under roadways/pavements. The Contractor shall provide evidence that the microduct Innerduct complies with industry standards for under roadways and ground burial similar to what is applied to EPC-40 polyvinyl chloride (PVC) (Schedule 40) in accordance with NEMA TC-2 or high-density polyethylene (HDPE) SDR 11.5. The ducts shall be appropriately labeled indicating the composition material. Ducts shall have a sleeve or bell end type coupling and shall be watertight when assembled.

Fiber Optic Cable

The Contractor shall install outside plant microfiber that meets industry standards and suitability for direct buried/bored applications with Tracer Wire. The intent is to install the cable in one continuous length, to the extent that it is practical.

MICROFIBER is required for this project. The Contractor shall provide the specifications as a submittal for review and approval by the AEDC White Oak Program Manager before purchasing.

The cable shall be installed along the recommended paths IAW attached drawings and diagrams Attachment 1. The cable shall meet Rural Utilities Service (RUS) 7 CFR 1755.900 criteria and shall comply with industry standards for:

- Manufacturers' cable marking
- Jacket and materials
- Rip cords
- Water blocking
- Fiber color coding
- Mode field diameter
- Core cladding concentricity
- Attenuation
- Dispersion characteristics at 1310nm and 1550nm, etc.

2.2.1.7.1.3 Installation.

New microduct innerduct shall not enter or exit handholes from the bottom floor but from the sidewalls.

2.2.1.7.1.4 Paved Surface Crossings. – N/A

2.2.1.7.1.5 Conduit Protection. If Applicable

The Contactor shall install the appropriate physical protection required (*i.e.* concrete encasement, steel tube casing) for buried Microduct, PVC conduits or HDPE roll pipe, consistent with commonly accepted telecommunications industry practices relative to the task and to the environment in which they are installed and the loads (H-5, H-10, H-20, railroads, flightline, etc.) to which they are expected to be exposed, with ground acidity factored. Type of protection may be dependent upon the specific application and shall be IAW applicable federal, state, local, or GSA Civil Engineering requirements. At a minimum, nonmetallic conduits shall be buried forty-eight inches (48") below the road surface and shall be pre-approved by Civil Engineering; otherwise, the conduit shall be required to be encased in concrete of minimum three-thousand pounds/per square inch ($3,000 \text{ lbs/in}^2$) compressive strength where vehicular traffic is above the pathway. The encasement shall be concrete of a wet type mix and shall be placed in such a manner as to ensure the concrete completely surrounds all conduits and no air voids are trapped in the mix. Concrete may not be required if buried below the surface deeper than forty-eight inches (48"), but final design and depth of conduit under roadways must receive prior approval from our GSA Civil Engineers.

2.2.1.7.1.6 Conduit Bends, Sealing and Mandrelling.

All bends between maintenance holes shall be a minimum forty-foot (40') radius with the sum of bends in all directions not exceeding a total of ninety degrees (90°). Only one ninety-degree (90°) bend shall be allowed between maintenance holes. Where a bend or sweep is placed in PVC nonmetallic duct bank between maintenance holes, the duct bank must be encased, as specified above in 2.2.1.7.1.5, in concrete with a minimum compressive strength of three-thousand pounds/per square inch ($3,000 \text{ lbs/in}^2$) as specified above in paragraph 2.2.1.7.5 for each CLIN. (See Attachment 1 below). Concrete encasement is not required between handholes not traversing below vehicle traffic roads or railroad tracks where duct is greater than thirty-nine inches (39") below the surface. All Duct less than thirty-nine inches (39") below the surface shall be encased in concrete. Ducts shall have bell ends and enter a maintenance hole perpendicular to the surface of the wall through which it is entering. All ducts/inner ducts entering maintenance holes must be sealed. Universal duct plugs or removable putty sealants may be used. Upon completion of conduit sections, a test mandrel one-quarter inch (.25") smaller than the inside diameter of the conduit shall be pulled through two diagonally opposite ducts to ensure proper alignment. In addition, all ducts shall be cleared of loose materials such as concrete, mud, dirt, stones, etc.

2.2.1.7.1.7 Building Entrances, Pull Boxes, and NEMA 4 Enclosures.

The Contractor shall serve building entrances with conduit or roll pipe installed within a trench and extended into the building. All conduit ends shall have connectors and bushings installed at any wall and/or pull box penetrations. Voids within conduits shall be sealed. Seams around conduit penetrations shall be sealed with the appropriate materials (silicon, firestop, or sealant) to meet applicable building codes to keep water from penetrating the building exterior surface.

Above ground building entrances (AGBE) shall consist of the following infrastructure attached to the building exterior wall:

- 1) Exposed, UV rated, galvanized steel conduits.
- 2) An exterior rated National Electrical Manufacturers Association (NEMA) 3 or better pull

box or galvanized LB type.

- 3) All mounting hardware be of the same finish, manufacturer, length, and installed at uniform distances.
- 4) Installation of a wall penetration and conduit sleeve through the building exterior wall.

Silicon shall be installed in the following points of the building's exterior surface:

- 1) All penetrations, including bolt holes and conduit.
- 2) All seams at surface mounted materials, including where exterior rated pull box meets the wall.
- 3) All materials identified in this section be constructed from a paintable material and be installed plumb, level, sealed, and with measures taken to ensure all work performed to the building's exterior surfaces is watertight.

In the installation of a new AGBE the quantity and size of the conduit or roll pipe entering the AGBE shall equal the amount leaving the AGBE and penetrating the building exterior wall.

A SBE shall be the preferred method, pending BAN design, of building entrance installation.

A SBE shall include:

- 1) Access to the lowest floor space immediately inside the building's exterior wall
- 2) The ability to core drill the floor of this space
- 3) The installation of conduit ends four inches (4") above finished floor (AFF)
- 4) Concrete to seal the space around the conduit penetration
- 5) The anchoring of conduit into place

2.2.1.7.1.8 Entrance Conduits into Existing Maintenance holes and Handholes

When new entrance conduits/ducts or sleeves are required, the Contractor shall bore and install the necessary holes and ducts or sleeves, if knockouts do not exist.

Penetration shall not be in locations through the wall blocking use of existing ducts in the handhole. New ducts will be a minimum of 18 inches from either the handhole floor or ceiling, if practical. The minimum bending radius for entry conduit/ducts shall be no less than ten times (10×) the inside diameter of the conduit. Ducts and openings around ducts shall be sealed to prevent moisture from entering the handholes. New duct shall not enter or exit handholes from the bottom floor but from the sidewalls IAW specifications provided.

2.2.1.7.1.9 Pull Tape. N/A

Since Microfiber will be blown through the Microduct, new Pull Tape will not be required.

2.2.1.7.1.10 Utility Separation.

When communications ducts cross either power duct or buried power cable, the Contractor shall maintain a minimum separation of three inches (3") of concrete or twelve inches (12") of well-tamped earth parallel or between the two. The Contractor shall maintain six inches (6") when crossing or twelve inches (12") when parallel for pipes (e.g., gas, water, oil).

2.2.1.7.1.11 Spacers and Tracer Wire.

Along the length of the duct run, if the ducts are installed by trenching, the Contractor shall install spacers at an interval of four (4) spacers per twenty feet (20') feet. The Contractor shall also bury cable warning tape one foot (1') below the surface following the duct route. The tape shall be a minimum of three inches (3") wide and orange in color with an appropriate warning message. All new tracer wire installed for this project shall have a wire nut and label installed at all wire ends, secured, but not connected to, grounds. The tracer wire shall be exposed, free from the conduit, and capped (insulated). The tracer shall be secured and routed to the handhole neck to a point accessible to maintenance personnel without having to enter the handhole and tagged with a label indicating it as a "Duct Tracer Wire to xxx - Do Not Remove", where xxx is the other end of the wire. Tracer wires shall not be connected to any grounding system. Tracer wire shall be pulled back from building entrances until it is underground to prevent lightning damage.

2.2.1.7.2 Closure for Underground Optical Fiber Cables.

Underground optical fiber cable closures shall be sealed and suitable for enclosing a splice organizer in a protective housing. The splice organizer shall consist of splice trays and protective sleeves, shield bond connectors, and ancillary hardware necessary to house fiber optic splices. The splice organizer shall house fiber optic fusion splices in a neat and orderly fashion. The splice organizer shall provide individual support for each splice. The organizer shall allow for thirty-nine inches (39") of fiber to be stored without kinks, twists, or micro-bends. The organizer shall be suitable for reentry, for future maintenance or modification, without damage to the fiber or splices. The closure shall be pre-formed high quality re-enterable splice case or equivalent for making a straight, butt, or branch splice and shall protect the splice and maintain cable shield electrical continuity in a handhole environment and suitable for pressure testing. The number of cable ports in and out of the closure, along with any necessary washers or any plugs for unused ports, shall be suitable for the application.

2.2.1.7.3 Fiber Optic Distribution Panels (FODPs).

The locations of the Cabinets or Fiber Optic Distribution Panels (FODPs) install location will be identified at the walk-through, see table below, FODPs shall be 4U.

2.2.1.8 Testing.

The Contractor shall furnish all test equipment and personnel required to conduct testing. The Contractor shall record all inspections and tests as they are accomplished and make all test sheets/results available for the AEDC White Oak Program Manager as tests are completed. The Contractor shall notify the AEDC White Oak Program Manager at least seven (7) calendar days prior to any testing. All testing will be IAW accepted telecommunications industry standards for the type of test being conducted. The Contractor shall provide test reports to the Government within ten (10) calendar days of completing testing. The Contractor shall locate/repair any testing irregularities caused during installation.

The Contractor shall correct and repair any splice errors detected. The Contractor shall deliver cable free from manufacturer defects, faults, or damage. The Contractor shall locate and repair cable faults or splicer errors identified during or after installation.

2.2.1.8.1 Fiber Optic Cable Tests.

The Contractor shall test all strands of all fiber optic cable in accordance with Telecommunications Industry Association (TIA) 526-7 Measurement of Optical Power Loss of Installed Single-mode Fiber Cable Plant, or equivalent. The Contractor shall, at a minimum, complete the following tests:

- Optical Time Domain Reflectometer (OTDR)
- Calibrated Light Source and Meter tests
- Optical Power Meter tests will be used for all end-to-end circuits
- Bi-directional testing at 1310nm and 1550nm is required between FODPs
- One-way OTDR testing for incomplete circuits ending in handholes

NOTE: Testing of the Fiber Optic Cables on the reel shall be provided to the AF Program Manager prior to installation.

2.2.1.8.1.1 Optical Attenuation.

The Contractor shall conduct end-to-end attenuation tests on all fiber optic cable strands to ensure the installed cable is within the specified parameters.

2.2.1.8.1.2 Fiber Testing.

The Contractor shall test all strands of fiber optic cable to determine the installed cable length between optical patch panels. All test results shall be provided as part of project completion documentation.

2.2.2 Site Restoration/Debris Removal.

The Contractor shall restore all disturbed surfaces to include backfilling, soil compacting, reseedling, re-sodding, or any other necessary material and services required to restore ground conditions to the original condition. Restoration at each location shall be subject to final inspection and approval by the Site Civil Engineer (BCE) or equivalent representative IAW guidelines set forth in the plans approved by the BCE representative. The Contractor shall perform follow-up grounds restoration as necessary due to surface settling, lack of turf germination, reseedling failure, etc.

The Contractor shall handle and dispose of all residues from this project IAW applicable laws and regulations. The AF Program Manager will coordinate the Host Base requirements for disposal. The Contractor shall notify the BCE representative of any contaminated soil discovered. The Contractor shall make every effort to minimize pollution and ensure proper disposal.

The Contractor shall restore all asphalt, concrete, brick, paving stone, etc. damaged in the performance of this Contract. Any asphalt, concrete, street, curb or sidewalk replacement shall be IAW with guidelines set forth by the BCE. At a minimum, the restoration shall match existing strength, color (to the extent practical), and type of material. The Contractor shall perform follow-up restoration if any location does not match its original condition.

2.2.3 Service Outages.

The Contractor shall work to prevent any unscheduled interruptions of communications capabilities at the site location. The Contractor shall coordinate planned outages with the AEDC White Oak Program Manager at least ten (10) calendar days in advance of the outage if the implementation necessitates disruption of service, (e.g., communications, electrical, or other utilities).

2.2.4 National Pollution Discharge Elimination System (NPDES).

Environmental coordination shall be accomplished through the multiple Environmental groups at White Oak MD. The Environmental POCs will be provided at time of kick off meeting

If required by the Civil Engineer Branch (AEDC/GSA), the Contractor shall provide all required application forms for State and/or Federal level to the Installation Management Section (AEDC/TSDCI) to obtain coverage under a General National Pollutant Discharge Elimination System (NPDES) Permit for Storm Water Discharges Associated with Construction Activity [e.g. Notice of Intent, Storm Water Pollution Prevention Plan (SWPPP), etc.].

The Contractor shall review the NPDES requirements applicable to the base, develop a plan, and comply with any other applicable State and/or Federal requirements for construction. The Contractor shall implement, monitor, and manage the NPDES Permit to ensure sediment caused by storm water runoff, does not enter storm drainage channels, and/or inlets.

The SWPPP is the sole responsibility of the Contractor. AEDC/TSDCI will review and advise to ensure minimum standards and storm water controls/best management practices (BMPs) are in place to ensure compliance.

2.2.5 Underground Utilities Markings.

The Contractor shall coordinate with through the AF and Host base POC/agencies to ensure markings are placed over existing base infrastructure prior to digging or directional drilling. The Contractor shall take precautions to protect existing infrastructure. The Contractor shall be responsible for costs associated with repair of any damage caused during installation where infrastructure is clearly marked. See section 2.1.4 for MWP Section IV-V (aka Dig Permit requests). The Contractor is responsible for maintaining all markings.

The Contractor shall clearly mark all Contractor-Furnished Property and Equipment (CFP/CFE) with their company's name. The Contractor shall place an easily read, visible sign (minimum 8.5" × 11") on large containers, equipment, or un-occupied rental vehicles while on FRC indicating the company name and both the Contractor and Site POC's names and local telephone numbers.

2.2.6 Installation Schedules.

The Contractor shall provide a complete milestone schedule denoting project activities including time-phased start and completion dates for the project and sub-projects associated with the installation of the components and system.

2.2.7 As-Built Drawings.

The Government will furnish existing system drawings, and the Contractor shall submit to the AEDC White Oak Program Manager for approval and acceptance, with the Contracting Officer included, updated drawings:

- **Civil Engineer (CE) drawings** in AutoCAD for buildings, facilities, site plans, pathways, etc.
- **Communication/Information Technology drawings** in Visio and .pdf format for rack elevation and Outside Plant (OSP). The OSP drawings shall depict the entire pathway and details of the installation, including but not limited to:
 - Labeling
 - Cables
 - Innerducts and maintenance holes/handholes
 - Conduits
 - Maintenance loops
 - Distances
 - Bores
 - Trenches
 - Building entrances
- **Detailed butterfly drawings** If the existing cable diagram or butterfly drawings are not available; the Contractor shall create the missing cable diagram/butterfly drawings and add the components used in this project.
- **All AF Civil Engineer (CE) type drawings shall (e.g. facility, site planning, pathways etc.):**
- **Final Design Drawings and Documentation:** Final design documentation including drawings, specifications, estimates, and other required deliverables. Electronic documentation submitted shall include every associated component file of drawings and other documentation, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files. Furnish a CD-ROM of all final design documentation, and drawings in AutoCAD 2007 through 2015 drawing file format (.dwg). The drawings shall be fully editable using AutoCAD software. Drawings shall be delivered to clearly show the location and arrangement of all materials and equipment. The only permissible text font is Romans, custom fonts and shape files are prohibited. The minimum allowable font size shall be one-eighth of an inch (0.125"). The maximum allowable font size shall be one quarter of an inch (0.25").
- **Record (As-Built) Drawings:** Electronic record drawings showing final configuration of work accomplished. Show all changes, additions, and deviations from the original contract drawings and documentation. If no changes occur, furnish certification to that effect. Drawings shall be redlined and shall accurately show as-built conditions during

the progress of the job. The Contractor shall deliver all drawings on a CD-ROM, in AutoCAD 2007 through 2015 format, including every associated component file of the drawings, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files.

2.2.8 Installation Test Plan.

The Contractor shall provide a test plan as to how the system shall be pre-tested, in-progress tested, and post-tested, along with a cut-over plan to demonstrate to the Government the system is fully operational and meets or exceeds the specified requirements and that the system is fully ready to be placed into service. The Contractor shall test the system to demonstrate its proper performance to the Government quality assurance representative. These tests shall be accomplished prior to the system being utilized. The Contractor shall submit an Acceptance/Installation Test Report of the results of the testing accomplished under the installation test plan. The Test Report(s) shall be provided to the AF Program Manager no-later-than ten (10) calendar days after test(s) have been completed.

2.2.9 Final Acceptance.

The Contractor shall schedule a final project walk-through with the AF Program Manager approximately ten (10) calendar days prior to acceptance.

2.2.11 As-Built Documentation in Custom Visualization Component (CVC)

The Contractor shall record geospatial data and provide as-built documentation (shape files) of all new installed maintenance hole system components (including metadata) compatible with the Cyberspace Infrastructure Planning System (CIPS) Custom Visualization Component (CVC) drawing system. Data points shall be recorded at the center of each handhole lid and at intervals not to exceed twenty-five feet (25') along cable routes. Sufficient data points shall be recorded to capture any change in direction along the route. All GPS coordinates shall have plus or minus three feet (+/-3') accuracy for all readings. The AF Program Manager will provide the Contractor with a copy of the installation's most current Base Wide Duct System drawings, and if available the GeoBase Common Installation Picture (CIP), and current CVC drawings of the areas of interest. The Contractor shall deliver shape files upon project completion.

2.2.12 Deliverable Summary.

All deliverables are subject to Government acceptance and approval and shall meet professional standards and the requirements set forth in this Contract. All deliverables shall be produced using recommended software tools/versions as specified by the Government.

3.0 GENERAL INFORMATION

3.1 Period of Performance

The period of performance for the project shall be determined based on the proposed schedule and actual Contract award date.

3.2 Place of Performance

The place of performance is US Army Adelphi Laboratory Center, Adelphi MD and USAF AEDC White Oak, White Oak MD.

3.3 Hours of Operation

The Contractor shall routinely work during normal duty hours. However, mission requirements may necessitate approximately 15% of the work outside normal hours (nights and/or weekends), especially if existing service must be interrupted. Any site work to be performed outside of normal duty hours shall be requested by the Contractor to the Contracting Officer for approval with the coordination of the AF Program Manager at least ten (10) calendar days in advance.

3.4 Holidays/Down Days

The Contractor shall not perform under this Contract on federal holidays or site-unique down-days without written authorization by the Contracting Officer and coordination of the AF Program Manager.

3.5 Base Support

On-base laydown areas are available. The Contractor shall propose and identify specific areas and sizes of any laydown area(s) in its proposal. Any other required base/Government support will be identified in a provided Project Support Agreement (PSA) for Government acceptance.

3.6 Payments

Payments will be made via Wide Area Workflow (WAWF) in accordance with the awarded Contract's line item numbers (CLINs) and clauses.

4.0 APPENDICES.

4.1 Appendix A. APPLICABLE STANDARDS.

- UFC 3-580-01 Telecommunications Interior Infrastructure Planning and Design, with Change 1 or latest change
- AFI 91-203 – Air Force Consolidated Occupational Safety Instruction OSHA CFR 29 Part 1910-268 – Telecommunications
- TIA-606-B - Administration Standard for Telecommunications InfrastructureTIA-568-C.0 - Generic Telecommunications Cabling for Customer Premises TIA-568-C.1 - Commercial Building Telecommunications Cabling Standard
- TIA-568-C.2 - Balanced Twisted-Pair Telecommunications Cabling and Components Standard TIA-568-C.3 - Optical Fiber Cabling Components Standard
- TIA 942 - Telecommunications Infrastructure Standard for Data Centers
- TIA-607-B - Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications
- TIA-526-7- Measurement of Optical Power Loss of Installed Single-mode Fiber Cable Plant TIA-526-14-B Optical Power Loss Measurement of Installed Multimode Fiber Cable Plant TIA-569-C - Commercial Building Standard for Telecommunications Pathways and Spaces TIA-570-C –Residential Telecommunications Infrastructure Standard
- TIA-598-C - Optical Fiber Cable Color Coding
- TIA-758-B Customer-owned Outside Plant Telecommunication Infrastructure Standard RUS Bulletin 1735F-401, Standards for Splicing Copper and Fiber Cable
- RUS Bulletin 1751F-643 - Underground Plant Design
- RUS Bulletin 1751F-801 – Electrical Protection Fundamentals
- RUS Bulletin 1753F-151 (515b) - Specifications and Drawings for Underground Cable Installation
- RUS Bulletin 1753F-201 (PC-4) – RUS Standard for Acceptance Tests and Measurements of Telecommunications Plant
- RUS Bulletin 1753F-207 (PE-87) – REA Specification for Terminating Cables NFPA 70 - National Electric Code